Mahesh Moholkar

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Summary

Software Engineer (3+ years) building production AI and full-stack products with **Python/FastAPI and Next.js**: a cited RAG helpdesk with MCP tools, LLM ticket triage, natural-language reporting and multi-agent systems, with evals in CI. Background in multi-tenant SaaS: a university Campus ERP (React, .NET) serving 8,000+ users.

Professional Experience

Software Engineer (Full-Stack & Applied AI) **Aug 2023 – Present** · *3 yrs 3 mos*
Shivam Enterprises *Pune, India*

- Shipped a cited **RAG helpdesk assistant** on **Python/FastAPI** that answers from policies and circulars, with ERP tools exposed over **MCP** and per-user access checks on personal data.
- Created **Campus Support**, a Next.js + FastAPI service desk where an LLM checks ticket completeness, classifies requests and suggests the assignee with confidence gating.
- Owned the **retrieval and evaluation pipeline**: hybrid search (pgvector + BM25, reciprocal-rank fusion, reranking) and golden-set evals with LLM-as-judge in CI; added natural-language report queries and document extraction to the ERP.
- Designed a **Campus ERP for a university and its affiliated colleges** on **React + .NET 10**, with colleges as tenants, serving **8,000+ users**; cut p95 under 400 ms at k6 peak load with Redis HybridCache.
- Engineered shared **auth** (JWT, role- and menu-based access control) and **notification** (outbox + Hangfire workers: email, SMS and push with retries, delivery tracking) **microservices** used by every ERP.
- Delivered separate **Admissions, Management and Procurement ERPs**, each with its own database, on the shared microservices.
- Set up Serilog + correlation IDs, OpenTelemetry/Grafana and **Azure DevOps CI/CD** with Testcontainers; wrote ADRs; mentored interns.

Stack: Python, FastAPI, Next.js, React, TypeScript, PostgreSQL, pgvector, Ollama, MCP, Redis, .NET 10, SQL Server, Docker, Azure DevOps

Frontend Engineer (Part-time, Remote) **Jan 2025 – Jun 2026** · *1 yr 6 mos*
Vision – Multi-tenant SaaS ERP for GCC SMEs *Kuwait*

- Built the **React 18 + TypeScript client** (Vite) for a multi-tenant GCC ERP – quotation, sales order, invoice, returns, purchase order and bill screens, inventory and VAT-aware totals – as typed forms (React Hook Form + Zod) with state-driven edit/post/void actions.
- Owned the client data layer: TanStack Router (loaders, search-param filters), TanStack Query with per-document cache invalidation and optimistic updates, Zustand for session state, real-time notifications over WebSockets.
- Shipped a bilingual **English/Arabic RTL-first UI** (i18next, shadcn/Radix + Tailwind design system), permission-manifest-driven navigation, printable document views, and a shared component library reused across the web app, public storefront and admin app.

Stack: React 18, TypeScript, Vite, TanStack Router, TanStack Query, Zustand, React Hook Form, Zod, Tailwind CSS, shadcn/Radix, i18next

Open Source Contributor (Zero to Merge Cohort) **Aug 2023 – Sep 2023** · *2 mos*
Cloud Native Computing Foundation (CNCF) *Remote*

- CNCF mentored open-source program (Incubator Badge): PRs to Kubernetes-ecosystem projects (Go); bug fix merged into **Ockam** (Rust); documentation contributions to Kubernetes SIGs and EvalML.

Projects

PagerSage – Multi-Agent Incident Response github.com/maheshmoholkar/pagersage

- Claude agents walk a FalkorDB graph of services, deploys and past incidents to find the root-cause change and page the owner.

Stack: Python, FastAPI, Claude (Anthropic SDK), FalkorDB, fastembed, Next.js, TypeScript, Docker

Echo – Embeddable AI Customer Support, Chat + Voice github.com/maheshmoholkar/echo

- PydanticAI agent answers chat and voice calls (Pipecat, Whisper, Kokoro) from a business's knowledge base, with human handoff.

Stack: Next.js, TypeScript, Python, FastAPI, PydanticAI, PostgreSQL, pgvector, Ollama, Pipecat, Better Auth

Vyavasay AI ERP – GST Billing & Inventory SaaS for Indian SMEs vyavasayapp.in

- Designed and built a multi-tenant GST ERP running in production: React web app, Next.js storefront, Expo Android app, .NET API.
- AI: Claude turns a supplier's bill photo into a draft purchase bill; the mobile app finds products by photo (Cohere + pgvector).

Stack: Next.js, React, TypeScript, TanStack Router, Expo, PostgreSQL, pgvector, .NET 10, EF Core, Hangfire, SignalR, Docker, Railway

Skills

AI/ML: Claude, OpenAI SDK, Ollama, PydanticAI, MCP, tool calling, structured outputs, hybrid RAG, reranking, LLM-as-judge evals
Backend: Python (FastAPI, Pydantic, SQLAlchemy), REST APIs (OpenAPI), JWT auth, RBAC, Go, C#/.NET 10 (EF Core, Hangfire)
Frontend: Next.js, React, TypeScript, TanStack Query, TanStack Router, Zustand, React Hook Form, Zod, Tailwind CSS, i18next, Expo
Data & Infra: PostgreSQL, pgvector, SQL Server, Redis, Docker, AWS, Azure DevOps, GitHub Actions, OpenTelemetry, Grafana
Testing: pytest, xUnit, Testcontainers, Vitest, Playwright, k6

Education

Dr. D. Y. Patil Institute of Management and Research **2022 – 2024**
Master's in Computer Application – CGPA: 8.0 / 10 *Pune, India*